Predicting Economic Damage of Natural Disasters

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Invalid Date

Data processing

Introduction

Background:

As global temperatures rise, climate-related disasters have increased in both frequency and severity over the past several decades (World Wild Life). Extreme weather events such as floods, droughts, hurricanes, etc. impose a huge human and real economic costs across the world. It is imperative that we understand the determinants of disaster severity (in terms of economic damages) for better policy planning and mitigative strategies.

The Emergency Events Database (EM-DAT), maintained by the Centre for Research on the Epidemiology of Disasters (CRED), keeps an internationally standardized data on natural diasters worldwide. The information includes disaster type, location, affected population, deaths, injuries, and economic damages.

Research Question:

How do different disaster characteristics (such as geographic location, time of year, type, and total casualties incurred) predict the total economic damage caused by natural disasters in USD?

- Specifically, we examine whether disaster type, location, timing, duration, and measures of human impact (deaths, injuries, and affected population) are associated with higher reported economic losses.

Motivation:

We were first drawn to this research question because were intrigued by this data; we were amazed by the granularity and the surprising breadth of information in this data. However, we realized that doing research on this matter could have real, practial uses.

First, identifying predictors of large economic damage is critical for both public and private institutions. Governments and international organizations allocate substantial resources towards mitigation and recovery efforts, yet the determinants of high-cost disasters vary across event types and regions. Same applies for insurance companies and hedge fund traders that try modeling key determinants relevant to a disaster for their comparative advavantage against peers in the market.

By analyzing global disaster data, this project aims to quantify which observable characteristics are most strongly associated with economic loss. We hope that these results may provide insight into which types of disasters and impact measures are most economically destructive.

Hypothesis Reasoning:

Based on economic reasoning and relevant research, we expect that…

1. Disasters with higher number of affected individuals (absolute number, number of deaths and injuries) will have a positive relationship with economic damage.
2. Certain disaster types will incur systematically higher economic damage than that of others.
3. The level of damage dollars will defer based on countries because of their respective infrastructure that exists to mitigate damage.

Data Description:

This data was obtained from the Emergency Events Database, otherwise known as EM-DAT, which is maintained by the Centre for Research on the Epidemiology of Disasters (CRED). The goal of the database is to compile standardized information on natural disasters that occur around the world. According to the information on metadata, an event is included in EM-DAT if one or more of the following criteria is met:

- 10+ people killed
- 100+ people affected
- Declaration of a state of emergency
- Solicitation of international assistance

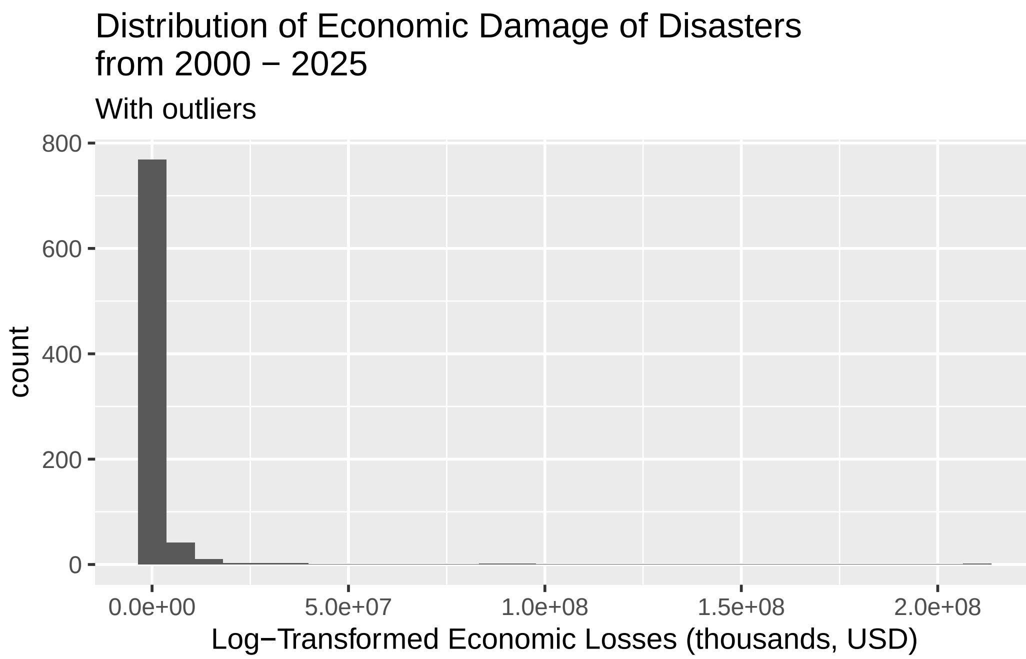
The observations describe different disasters that occurred around the world. Of each disaster, data on the number of people injured, affected, homeless, or killed, the total economic damages, type of disaster, exact location, start date and end date of the disaster were collected.

The data dictionary can be found here.

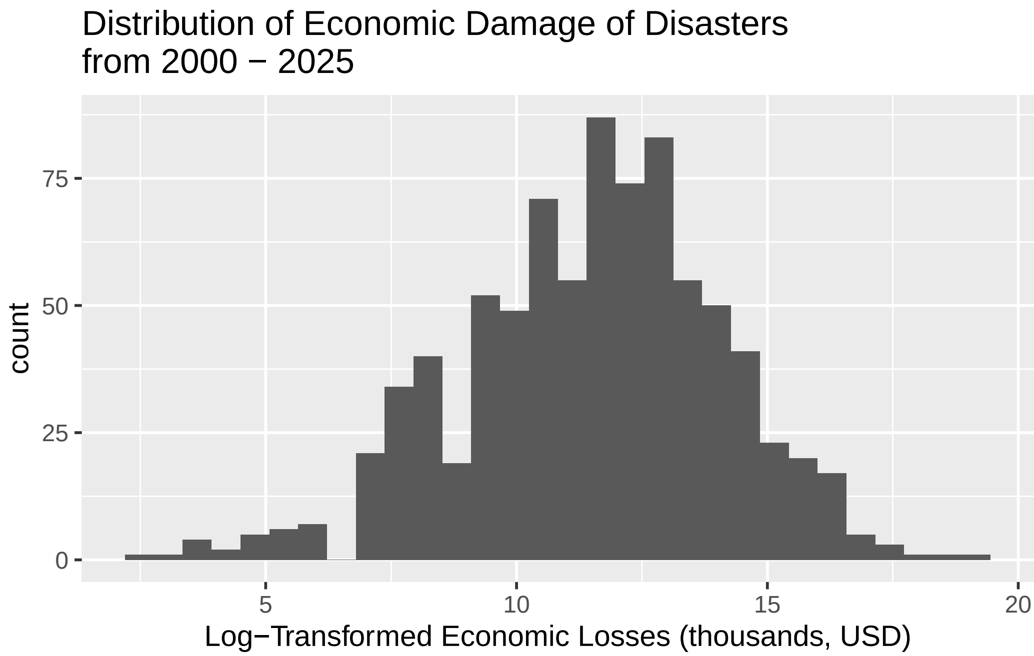
Limitations and Concerns

There are a considerable number of outliers in our data set with respect to total economic damage, duration in days, and many other variables. This motivated us to apply the logarithmic function to those variables. These outliers, however, should still be considered in the larger-picture analysis.

Exploratory Data Analysis




```
 Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
   12     20000    109576   1519107    559750 210000000 
```




```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  2.485   9.903  11.604  11.493  13.235  19.163 
```

```
[1] 2.606016
```

The distribution of the unchanged response variable is right-skewed and unimodal. The center is best represented by the median, which is $109,576,000, and the spread is best represented with the IQR, which is $539,750,000 (559,750,000 - 20,000,000). There are 123 (828-705) outliers present in the upper end of our distribution.

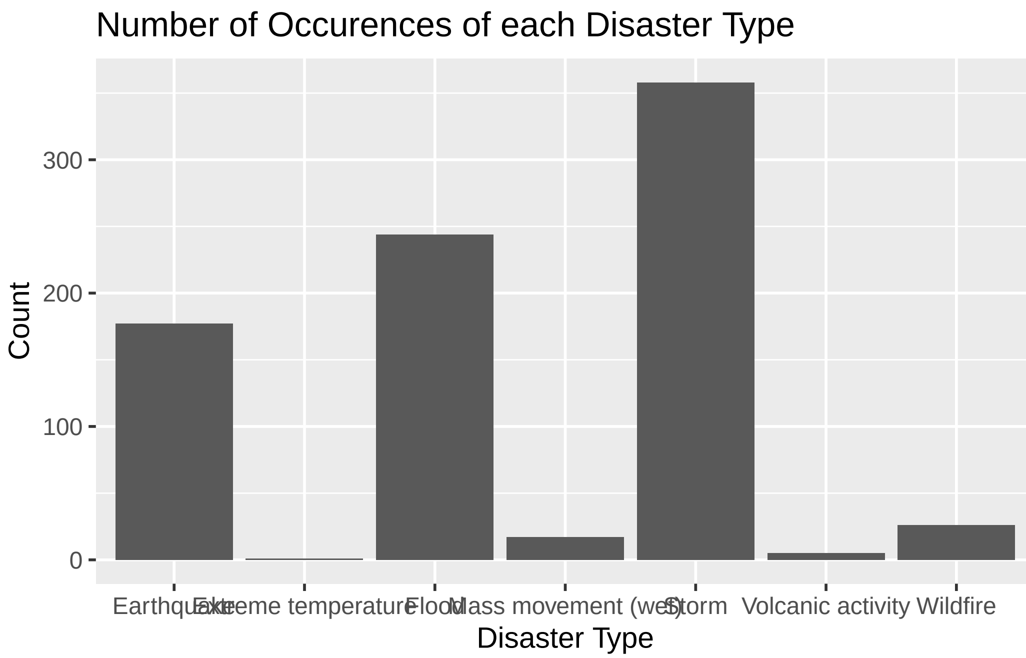
The distribution of the logged response variable is unimodal and symmetric. The center is best represented by the mean, which is $98,027,173.52 (e^11.493 * 1000), and the spread is best represented with the standard deviation, which is $13544.98 (e^2.606016 * 1000). There are outliers present in our distribution. **We will be working with the logged response variable**, log of total economic damage in thousands of USD, because without the log, the outliers skew the data heavily.

Predictors of interests:

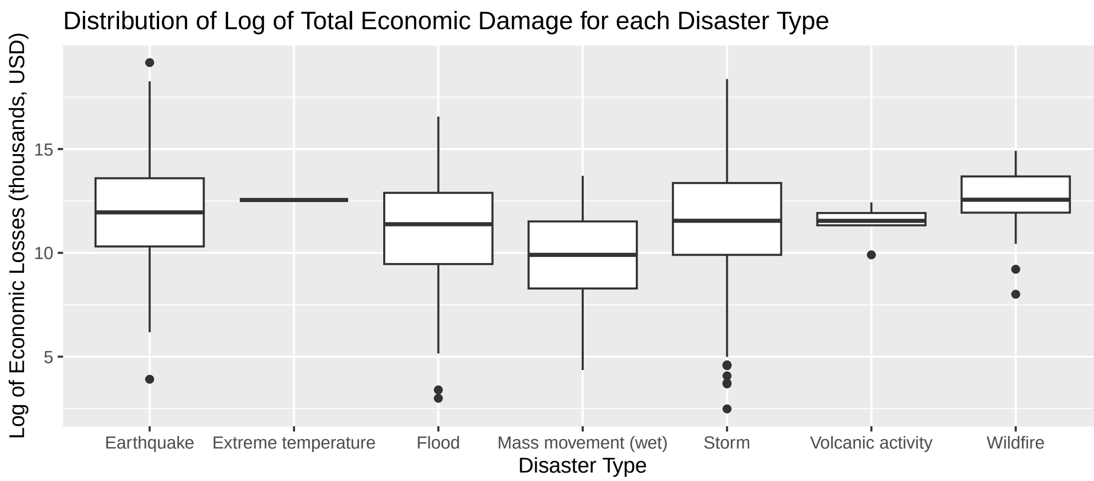
1. disaster_type - the type of disaster (categorical)

```
# A tibble: 7 x 1
  disaster_type
```

```
  <chr>
1 Flood
2 Storm
3 Earthquake
4 Mass movement (wet)
5 Wildfire
6 Volcanic activity
7 Extreme temperature
```



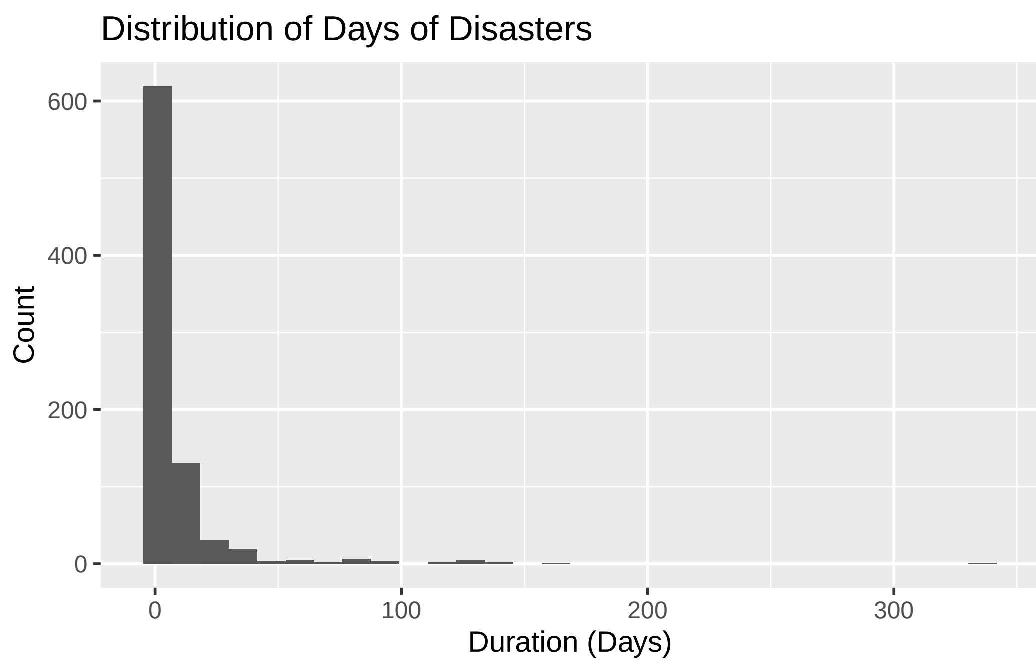
Storms, floods, and earthquakes are the most common disaster types, composing most of the data. Storms are the most common. Extreme temperature, wet mass movements, volcanic activity, and wildfires are the least common disaster types, with extreme temperature being the least common.

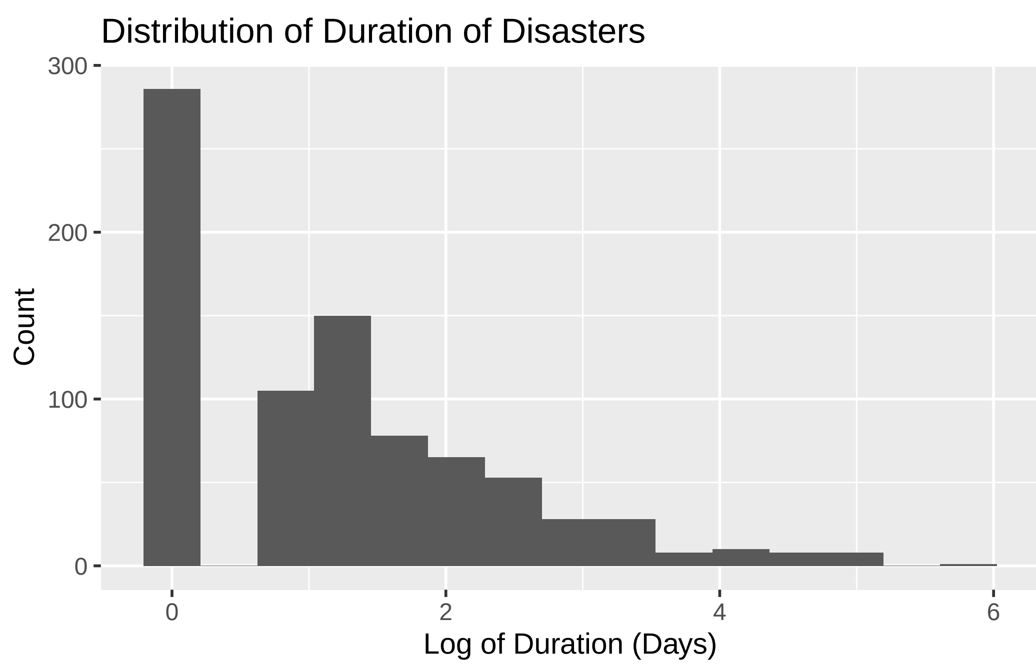



```
# A tibble: 7 x 3
  disaster_type       median   IQR
  <chr>                <dbl> <dbl>
1 Earthquake           12.0  3.28
2 Extreme temperature  12.5  0
3 Flood                11.4  3.43
4 Mass movement (wet)   9.90 3.23
5 Storm                11.5  3.46
6 Volcanic activity    11.5  0.592
7 Wildfire             12.6  1.75
```

Wildfires have the highest median log of total economic damage (in thousands of dollars), which is 12.558857, and thus, normal total economic damage, which would be 284604.84 thousands of dollars (e^12.558857). The median of earthquakes is slightly lower, and then the median of volcanic activities, floods, and storms, which all have similar values, slightly below the median of earthquakes. Wet mass movements have the lowest median (log of) total economic damage. Storms, floods, and earthquakes have the highest variability in the (log of) total economic damage.

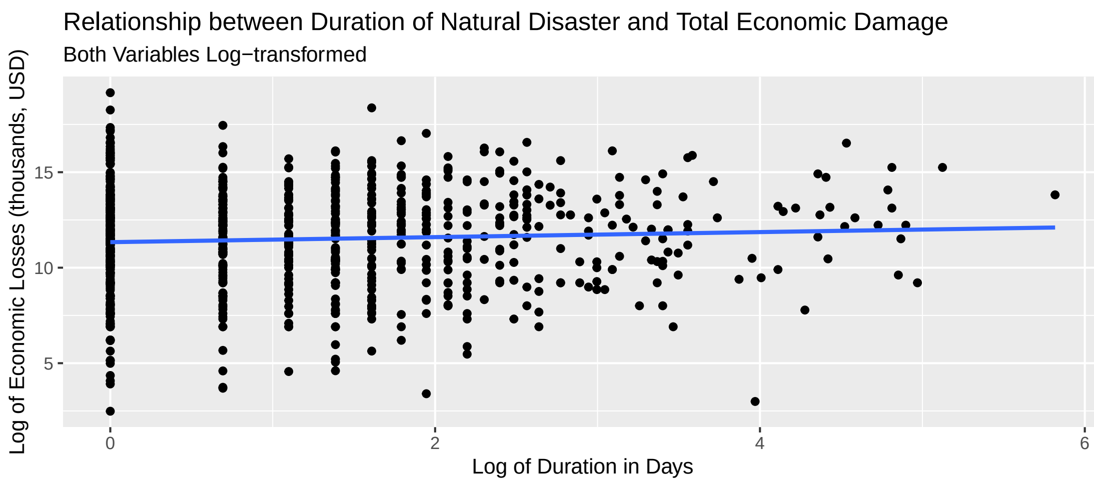
2. duration - duration in days of the disaster (quantitative)





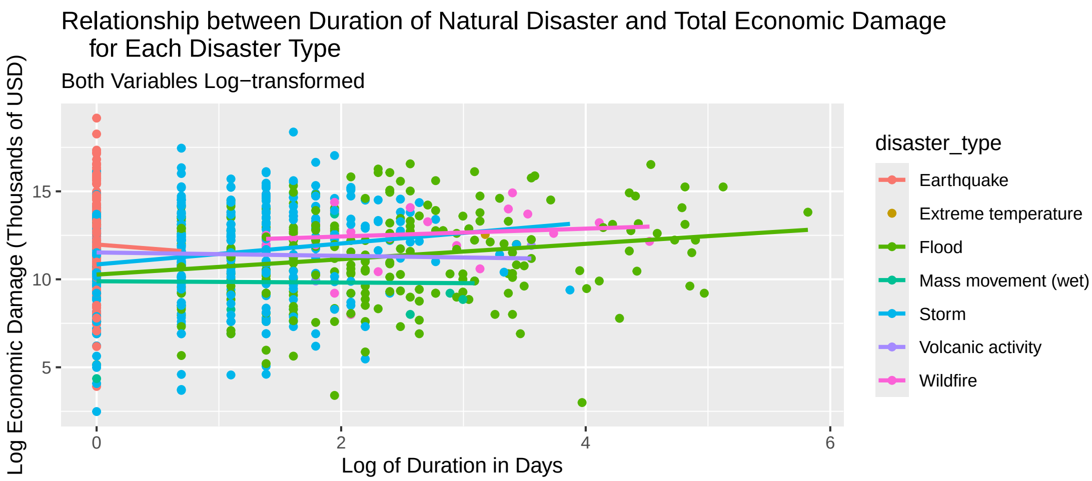

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   0.000   1.099   1.187   1.946   5.817
```

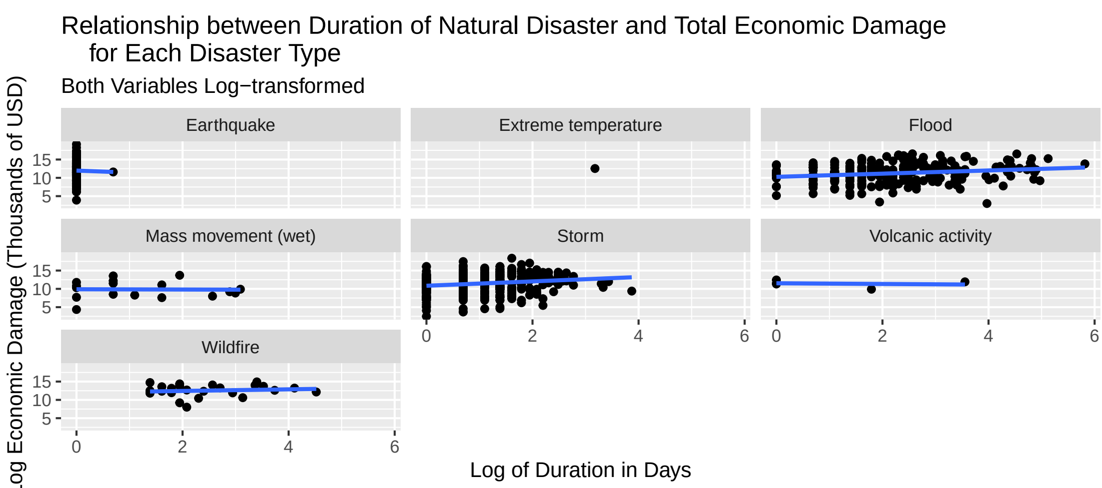
Because of the extreme outliers and extreme skew in the histogram of the duration (in days) of each disaster, we decided to explore the log of the duration of the disaster. The histogram of the log of the duration has right skew and is bimodal. This makes sense because a lot of the disasters only last 1 day long due to their inherent nature, such as earthquakes. The median is 1.099 log duration in days, and the IQR is 1.946 log duration in days. There are outliers in the upper end of our distribution.




```
# A tibble: 1 x 1
  r_squared
      <dbl>
1   0.00360
```

There seems to be a weak positive linear relationship between the log of duration in days with log of total economic damage (in thousands of dollars). The r-squared value is 0.003596655. There is a lot of variability in the response variable when the log of duration in days is low, but the variability decreases as the x value increases. There are outliers present throughout the graph.



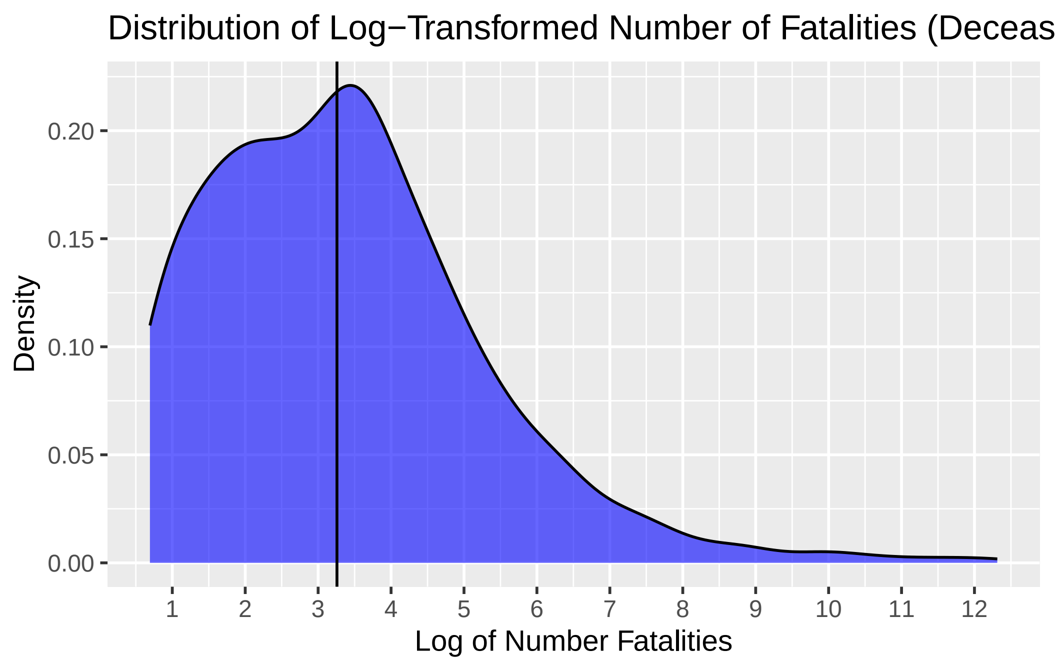


After visualizing the potential interaction effect between log duration of days and type of disaster, it is evident that an increase in the log duration in days has a stronger effect for storms and floods, compared to wildfires, volcanic activities, and wet mass movements, which is shown by the slightly steeper positive slope. This makes logical sense as well. We will consider putting this interaction effect into our model because of this visualization.

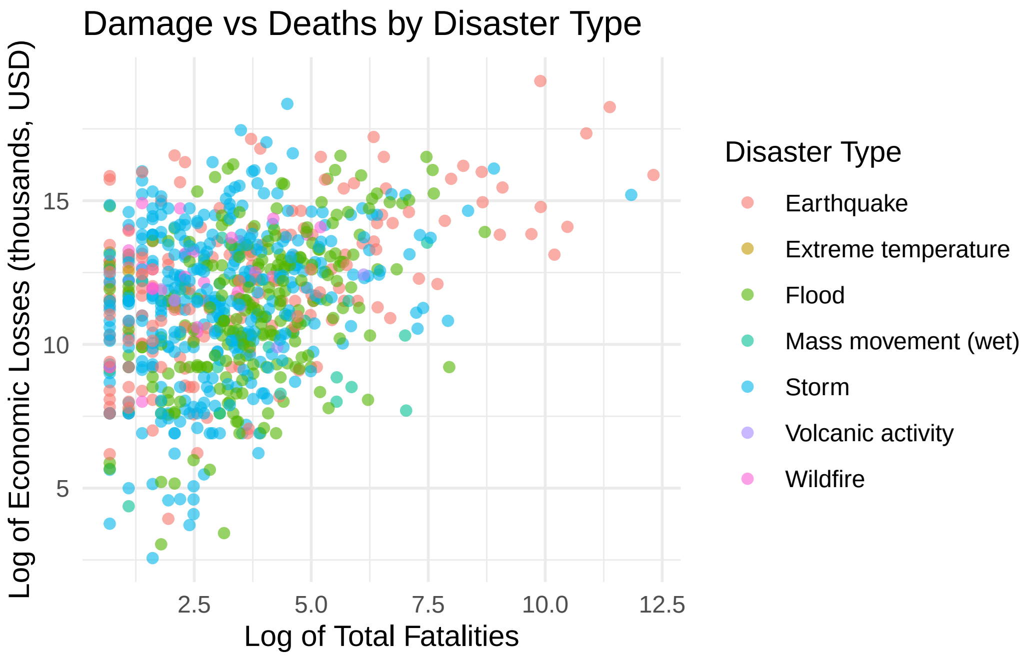
3. total_deaths - Total fatalities (deceased and missing combined) (quantitative)

Statistic	Value
Min.	1.000
1st Qu.	6.000
Median	25.000
Mean	921.675
3rd Qu.	80.000

Statistic	Value
Max.	222570.000



Because of the extreme skewness and outliers in the data of total fatalities (of each natural disaster), we decided to explore a log transformation of the data. In the density plot above, we can see that the overall distribution is unimodal and has a distinct right skew. From the summary statistics, we can tell that while the median number of fatalities is 25 people, some grand disasters have a catastrophic number of deaths, with the highest number being 222570 people.

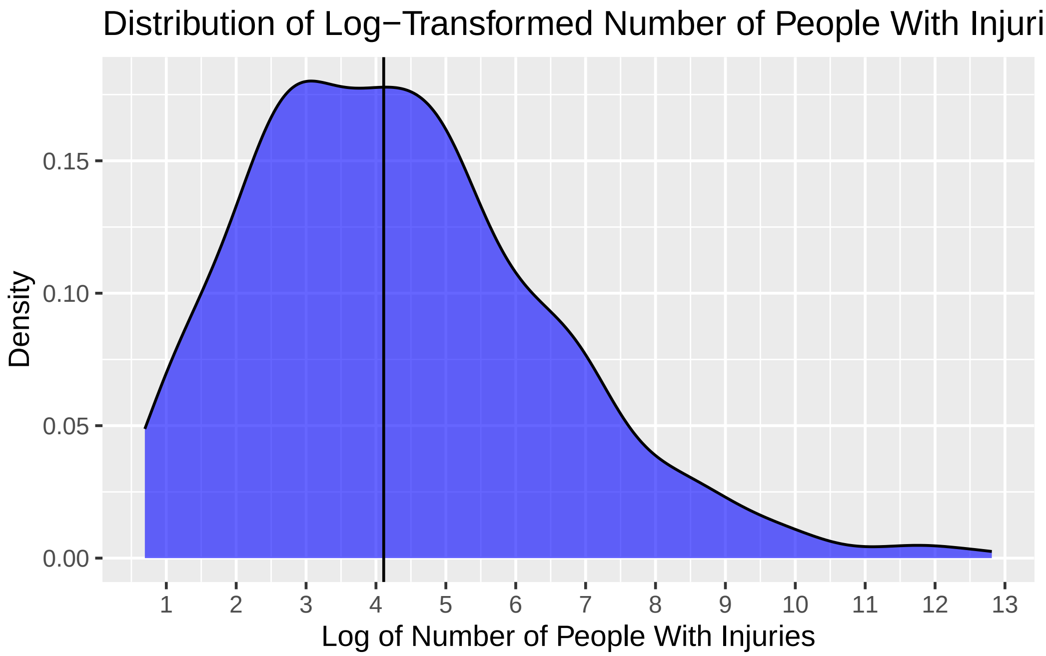


There is a clear positive relationship between total deaths and economic damage. Disasters with higher fatalities generally correspond to higher economic losses (USD). However, the relationship is noisy and dispersed, indicating that deaths alone may not be a strong predictor of damage and that further analysis needs to be done.

Differences across disaster types are visible. Earthquakes and storms tend to generate higher economic damage at comparable fatality levels, while floods and mass movements appear more concentrated at lower damage levels. This suggests that the impact of fatalities on economic damage likely varies by disaster type. This indicates potential interaction effects are present.

4. no_injured - Number of people with physical injuries, trauma, or illness requiring immediate medical assistance due to the disaster (quantitative)

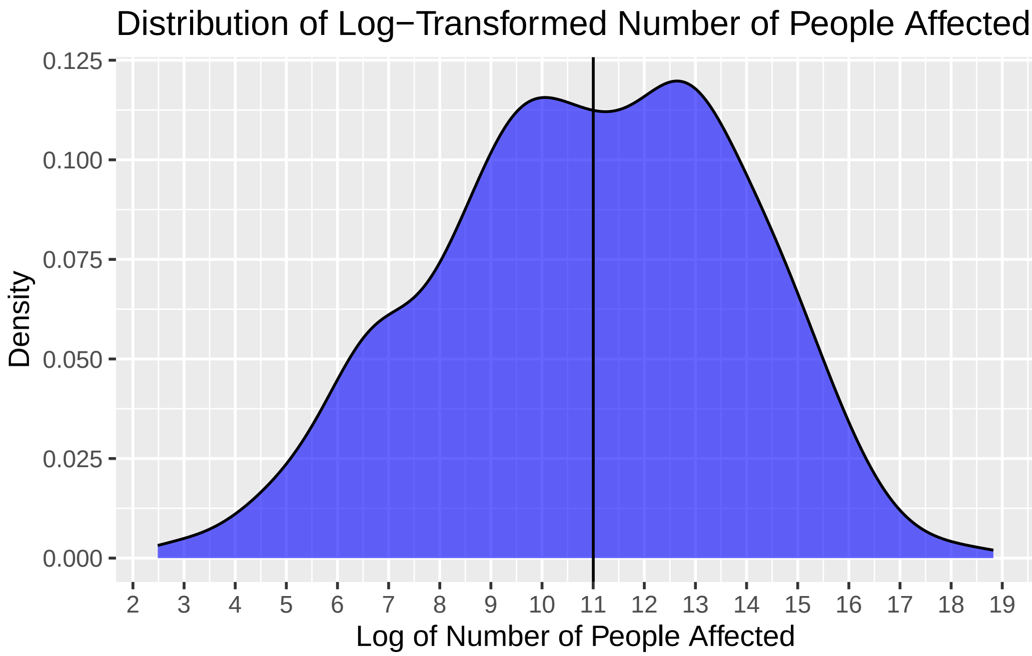
Statistic	Value
Min.	1.00
1st Qu.	14.00
Median	60.00
Mean	2338.36
3rd Qu.	266.25
Max.	366596.00



The story is similar to that of the previous predictor, total_deaths. There is a clear unimodal shape with a right skew, and we know that this skewness has been alleviated by the log transformation (so in reality, much larger). It makes sense that the median is higher at 60 people because, common-sensically, there will always be more people injured than killed at any given natural disaster. This corresponds to the maximum number being higher than that of total_deaths at 366596 people.

5. no_affected - Number of people requiring immediate assistance due to the disaster (quantitative)

Statistic	Value
Min.	11
1st Qu.	7000
Median	60000
Mean	1325411
3rd Qu.	500250
Max.	150000000



We again applied a log transformation to better visualize the distribution. In the density plot above, the distribution of the log-transformed values appears roughly unimodal, though there is still a right skew, considering the distribution before the log transformation. Compared to fatalities, the distribution of affected individuals is more spread out, reflecting the wide range of disaster scales – from local events to those impacting entire countries.

From the summary statistics, there is a gap between the median (60,000 people) and the mean (over 1.3 million people), indicating a big right tail driven by a small number of extremely large disasters. The maximum value of 150,000,000 people affected highlights the scale at which certain disasters can disrupt entire populations. This makes sense considering the nature of natural disasters: while many events are relatively small, large-scale disasters can impact vast populations through displacement and infrastructure damage.

> **!** Important
>
> Before you submit, make sure your code chunks are turned off with `echo: false` and there are no warnings or messages with `warning: false` and `message: false` in the YAML.